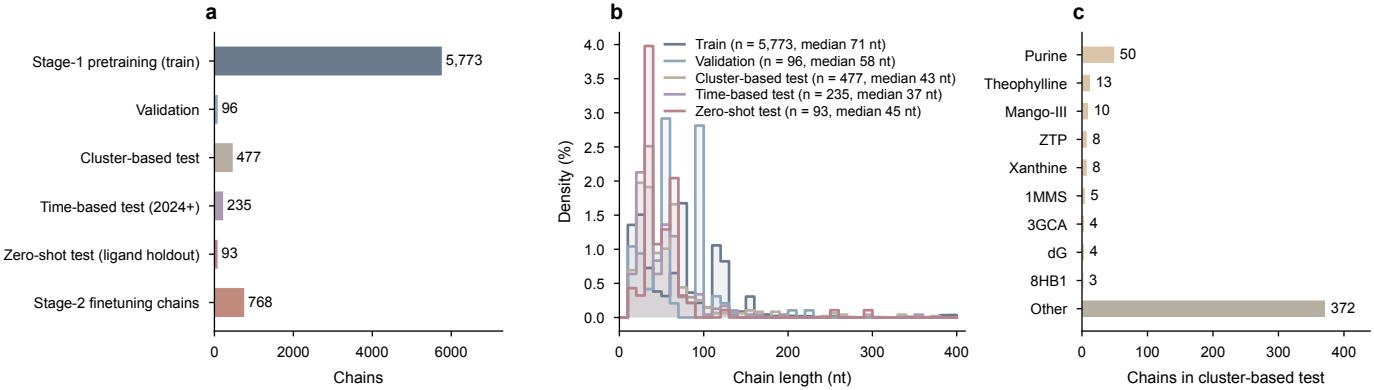
a
Stage-1 pretraining (train)
5,773
Validation
96
Cluster-based test
477
Time-based test (2024+)
235
Zero-shot test (ligand holdout)
93
Stage-2 finetuning chains
768
0
2000
4000
6000
Chains
b
Train (n = 5,773, median 71 nt)
Validation (n = 96, median 58 nt)
Cluster-based test (n = 477, median 43 nt)
Time-based test (n = 235, median 37 nt)
Zero-shot test (n = 93, median 45 nt)
Density (%)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
100
200
300
400
Chain length (nt)
c
Purine
50
Theophylline
13
Mango-III
10
ZTP
8
Xanthine
8
1MMS
5
3GCA
4
dG
4
8HB1
3
Other
372
0
100
200
300
400
Chains in cluster-based test